

ULTRA LOW-FIELD TO HIGH-FIELD MRI TRANSLATION USING ADVERSARIAL DIFFUSION

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ABSTRACT

Ultra Low-field Magnetic Resonance Imaging (MRI) scanners can potentially make a substantial impact in the field of medical imaging and radiology due to their cost-effectiveness, potential for portability and utility in an environment where the resource is in shortage. However, low-field MRI encounters challenges such as a low signal-to-noise ratio which results in lower-quality images. In this study, we introduce a novel image translation technique that relies on an adversarial diffusion-based deep learning approach to generate high-field MRI images from ultra low-field MR images. We have integrated a non-diffusive attention-guided module to enhance areas recognized as critical high-level features using self-attention maps from the diffusion process. To evaluate our approach, we use paired datasets consisting of different MRI sequences from both 64mT ultra low-field and 3T high-field scanners. We compare the performance of our method against state-of-the-art GAN and diffusion-based models, demonstrating its superior performance both quantitatively and qualitatively.

Index Terms— Image-to-image translation, Ultra Low-field MRI, Diffusion, Adversarial, Synthesis

1. INTRODUCTION

Magnetic Resonance Imaging (MRI) is now widely applied in clinical diagnosis due to its non-invasiveness, with a strong emphasis on high-field (HF) scanners operating at 1.5T to 3T. Nevertheless, in many middle and lower-income countries, Ultra Low-Field (ULF) MRI scanners are preferred due to their cost-effectiveness, ease of installation, and adaptability to power limitations [1]. Furthermore, these ULF MRI scanners are particularly valuable for patients with implanted medical devices, as they prevent exposure to potentially harmful high electromagnetic fields [2].

However, ULF MRI scanners suffer from certain limitations due to lower signal-to-noise ratio (SNR), resulting in low-resolution images that provide less diagnostic information compared to HF scans. This reduction in image quality can lead to inaccuracies in diagnoses, which impacts their

clinical utility. Moreover, in specific clinical scenarios [3], the presence of high-quality MR images is essential for an accurate diagnosis. Consequently, several novel image translation techniques now focus on synthesizing HF MR images from their ULF contrasts.

With the recent advancements in the application of deep learning techniques to medical image synthesis, a variety of deep neural networks have been employed to enhance the quality of ULF MRI images, including CNNs, U-Nets, and GANs [3, 2]. The majority of these methods have primarily focused on frameworks such as Image Quality Transfer (IQT), resulting in improving the quality of the ULF images [1]. Moreover, several super-resolution and multi-contrast synthesis methods have demonstrated their ability to synthesize HF MR images accurately [4, 2, 5].

However, most of the existing HF MRI synthesis methods relying on conventional CNN-based architectures and advanced networks like diffusion-based models remained underexplored [6, 7]. Moreover, the majority of existing approaches generate ULF image datasets through downsampling or simulating from HF images, lacking actual ULF data distribution and generalizability. Furthermore, many of these techniques focus primarily on mapping high-level spatial features, often neglecting specific regions that require greater attention and refinement of synthesized images to resemble HF MR images closely.

To address the above issues comprehensively, we propose a novel adversarial diffusion-based approach for ULF to HF MRI translation on actual data. Moreover, we incorporate a non-diffusive attention-guided module to generate high-quality images that closely resemble the ground truth by enhancing the informative features. To the best of our knowledge, this is the first diffusion-based approach for synthesizing HF MR images from ULF MRI data. We showcase the efficacy of this approach across various MRI contrasts.

2. METHODOLOGY

We propose a diffusion-based approach for ULF to HF MRI translation. The framework comprises a 2D adversarial diffusion-based translation module and a non-diffusive attention-guided refinement module.

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2.1. Denoising Diffusion Probabilistic Models (DDPMs)

DDPMs [6] represents a parameterized Markov chain that is trained to learn the transformation from pure noise to actual data over a finite number of time steps denoted as T . DDPMs consist of two primary processes: the forward and reverse processes. In the forward process, eqs. (1) and (2), random Gaussian noise is added to the input image across a sufficiently large number of time steps to produce the noisy image x_t .

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}x_{t-1}, \beta_t I) \quad (1)$$

$$x_t = \sqrt{1 - \beta_t}x_{t-1} + \epsilon\sqrt{\beta_t}; \epsilon \sim \mathcal{N}(0, I) \quad (2)$$

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sum_\theta(x_t, t)) \quad (3)$$

where ϵ is the added noise, β_t is noise variance, $\mathcal{N}$ is the Gaussian distribution and I is the identity covariance matrix.

Each reverse step within the diffusion model is mapped through a neural network, eq. (3) and trained by optimizing a lower bound on a simplified log-likelihood. Furthermore, without making a direct prediction of x_{t-1} , the DDPMs are considered parametrized models as in eq. (4),

$$p_\theta(x_{t-1}|x_t) \sim q(x_{t-1}|x_t, x_0 = f_\theta(x_t, t)) \quad (4)$$

where x_0 is predicted using the denoising model, $f_\theta(x_t, t)$, and then x_{t-1} using posterior distribution $q(x_{t-1}|x_t, x_0)$ [6].

2.2. Adversarial Diffusive Module

To synthesize HF MR images from ULF MRI contrasts, we employ an adversarial diffusion-based approach [7] where denoising distribution is modelled with a conditional GAN (Fig. 1). Compared to conventional diffusion models which utilize relatively smaller step sizes to perceive larger time steps (T), this model enables denoising for larger step sizes for fast diffusion. Similar to the forward diffusion process (Fig. 1, P_A) in conventional denoising diffusion, we first added noise to the target image x with a smaller time step ($T = 4$). In the reverse diffusion process (Fig. 1, P_B), we condition the denoising process with ULF MRI (y). We use a conditional GAN, G_θ , for denoising to estimate $p_\theta(x_{t-1}|x_t, y)$ as follows,

$$p_\theta(x_{t-1}|x_t, y) \sim q(x_{t-1}|x_t, \tilde{x}_0 = G_\theta(x_t, y, z, t)) \quad (5)$$

Then, G_θ is trained using the non-saturating adversarial loss [8] and L_1 loss given as below,

$$\begin{aligned} \max_\theta \sum_{t \geq 1} \mathbb{E}_{p_\theta(x_{t-1}|x_t, y)} [-\log(D_\theta(\tilde{x}_{t-1}, x_t, t))] \\ + \lambda L_1(G_\theta(x_t, y, t, z), x) \end{aligned} \quad (6)$$

where $\tilde{x}_{t-1}$ is the estimated denoised step of x_{t-1} and z is the random latent vector. A time-dependent discriminator D_θ is used to distinguish x_{t-1} and x_t by deciding if x_{t-1} is a plausible denoised version of x_t . The discriminator D_θ is trained by minimizing the following objective function,

$$\begin{aligned} \min_\theta \sum_{t \geq 1} \mathbb{E}_{q(x_t)} [\mathbb{E}_{q(x_{t-1}|x_t)} [-\log(D_\theta(x_{t-1}, x_t, t))] \\ + \mathbb{E}_{p_\theta(x_{t-1}|x_t, y)} [-\log(D_\theta(\tilde{x}_{t-1}, x_t, t))] \end{aligned} \quad (7)$$

2.3. Attention Guided Non-diffusive Module

Self-attention mechanism [9] plays a crucial role in diffusion models, allowing them to focus on finer and more significant aspects of images during the denoising process. Leveraging these important attributes, we introduce a non-diffusive module (Fig. 1, P_C) guided by attention maps to enhance the crucial high-level features. In this phase, we also aim to tackle the challenge of generating HF images that closely resemble acquired real images. This approach is essential because GANs tend to produce diverse image samples, potentially neglecting refined details of both global and local features and introducing artifacts in the absence of additional constraints.

First, we synthesize HF contrast for a given ULF conditioned image using diffusive generator G_θ . This synthesized image serves as input for our attention-guided module. Then, to enhance informative features from the hypointense brain regions within the ULF conditioned image, we extract finer details, such as edge features, to make the synthesized image visually consistent across all regions. To achieve this, we normalize the ULF image to a $[0, 1]$ range (Y_A) and use the inverse $(1 - Y_A)$ transformation as an attention map on the synthesized HF image. Then, we derive the aggregated self-attention maps (SA) from the G_θ and employed global average pooling (GAP) to combine the stacked self-attention maps from the middle and decoder layers (eq. 8).

$$SA_l = \text{Upsample}(GAP(\text{softmax}(Q_l K_l^T))) \quad (8)$$

where SA_l is the l^{th} layer attention map from and Q_l, K_l are features projected into queries and keys in the self-attention block [9]. Then we utilized nearest-neighbor upsampling to align the resolution of these attention-enhanced feature maps with the output image feature map obtained through a convolutional layer (conv).

Subsequently, these attention-weighted feature maps undergo a squeeze-and-excitation block (SE Net), consisting of two fully connected layers with SiLU and Sigmoid activations. This process derives weighted feature maps that reflect the importance of global feature compression. Finally, these weighted feature maps are passed through a simple encoder-decoder module (R_θ) trained using L_1 loss between synthesized and real HF images to enhance the results by eliminating any undesirable artifacts.

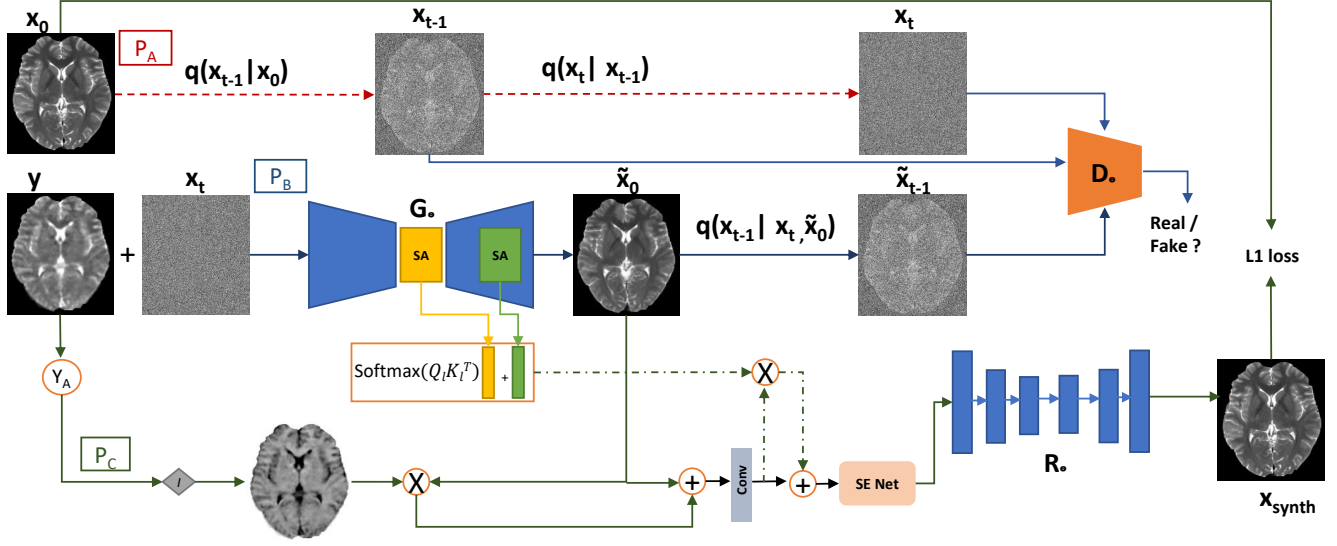


Fig. 1. Overview of the proposed architecture consists of Diffusive Module (with Forward Diffusion Process (P_A), Reverse Diffusion Process (P_B)), and Attention-guided Non-Diffusive Process (P_C).

2.4. Architectural details

The G_θ follows the U-Net architecture framework [7], featuring a series of resnet blocks and attention blocks [9]. Additionally, we incorporate sinusoidal positioning embeddings for time-step conditioning. Meanwhile, the D_θ is structured as a convolutional network with ResNet components [7]. The R_θ also comprises resnet blocks, similar to G_θ .

The hyperparameters of the diffusive module are as follows: Adam optimizer ($\beta_1=0.5$, $\beta_2=0.9$), learning rate: 0.001, number of diffusive steps:4, noise variance range: $\beta_{min}=0.1$ to $\beta_{max}=20$. Both modules were trained for 40 epochs in PyTorch on a single NVIDIA A40 graphics processing unit with 46 GB of memory.

3. EXPERIMENTS AND RESULTS

3.1. Dataset and Baselines

All the experiments for collecting ULF and HF MR images were conducted at Monash Biomedical Imaging, utilizing both the Hyperfine Swoop (64mT) and Siemens Biograph mMR 3T imaging systems. Data preprocessing involved resampling the data using *SynthSeg*⁺ [10] and applying bias correction through FSL-FAST. The paired ULF and HF images were obtained by performing co-registration using FMRIB's linear image registration tool [11]. To assess the proposed method, we selected 100 middle axial slices out of 160, resized to 256×256 , from 45 random healthy subjects: 20 for training, 10 for validation, and 15 for testing. Each subject contributed T1-w, T2-w, and fluid-attenuated inversion recovery (FLAIR) contrasts. We compared our proposed model against four state-of-the-art GAN and diffusion-based

image synthesis models including pix2pix [12], pGAN [13], GAN model with attention generator [14] (AttGAN) and DDPM [6].

3.2. Results

We assessed our proposed model for all three MRI contrasts via peak signal-to-noise ratio (PSNR), structural similarity index (SSIM) metrics and Mean Absolute Error (MAE). The results in Table 1 show that our model outperforms all other state-of-the-art image synthesis models up to 2.59% dB on PSNR and 19.4% on SSIM.

Table 1. Performance comparison for synthesis of HF T1,T2 and FLAIR contrasts (mean $\pm$ std.).

	T1			T2			Flair		
	PSNR (dB)	SSIM	MAE	PSNR (dB)	SSIM	MAE	PSNR (dB)	SSIM	MAE
pix2pix	23.439 ± 1.659	0.849 ± 0.046	0.027 ± 0.008	23.959 ± 1.512	0.885 ± 0.028	0.023 ± 0.005	22.273 ± 1.474	0.837 ± 0.046	0.030 ± 0.008
pGAN	22.937 ± 1.598	0.866 ± 0.033	0.029 ± 0.009	24.935 ± 1.157	0.905 ± 0.025	0.020 ± 0.004	22.668 ± 1.589	0.854 ± 0.041	0.028 ± 0.008
AttGAN	23.578 ± 1.484	0.818 ± 0.084	0.056 ± 0.013	24.356 ± 1.636	0.900 ± 0.031	0.082 ± 0.016	23.036 ± 1.370	0.852 ± 0.067	0.075 ± 0.014
DDPM	22.479 ± 1.575	0.712 ± 0.056	0.036 ± 0.008	25.433 ± 1.268	0.864 ± 0.030	0.022 ± 0.003	22.457 ± 1.501	0.758 ± 0.043	0.033 ± 0.007
Ours	24.039 ± 1.870	0.907 ± 0.030	0.024 ± 0.008	26.551 ± 1.517	0.933 ± 0.020	0.017 ± 0.004	23.942 ± 1.656	0.888 ± 0.034	0.023 ± 0.007

To evaluate our proposed method, we compared with other literature methods in Fig. 2. The visual comparison of the synthesized results clearly demonstrates that the proposed method can synthesize HF images that closely resemble the acquired images. In contrast, other models produce images with noticeable noise and inaccuracies in tissue contrasts and anatomical details.

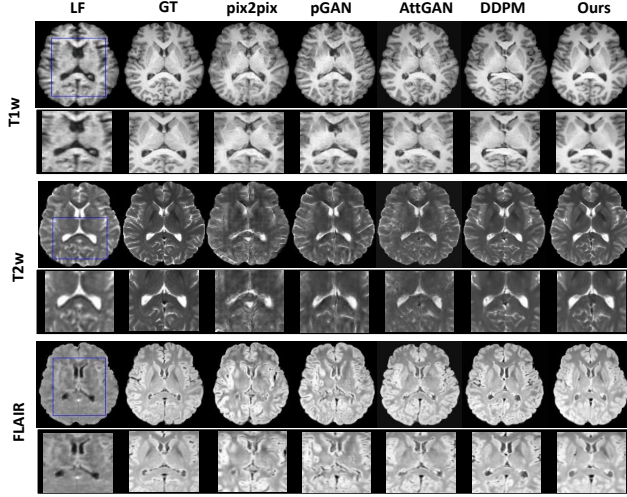


Fig. 2. Comparison of the synthesized HF MRI contrasts.

3.3. Ablation Study

We further conducted an ablation study for our non-diffusive attention-guided module and the adversarial loss component of the diffusive model. The results presented in Table 2 show the effectiveness of the attention-guided module in achieving accurate image synthesis and Frechet Inception Distance (FID) scores in Table 3, reveals the importance of adversarial loss in the diffusive module.

Table 2. Ablation of attention guided non-diffusive module.

	T1		T2		Flair	
	PSNR(dB)	SSIM	PSNR(dB)	SSIM	PSNR(dB)	SSIM
w/o Atten. guided non-diffusive module	23.990 ±1.809	0.902 ±0.031	26.440 ±1.600	0.930 ±0.020	23.770 ±1.590	0.880 ±0.030
Ours	24.039 ±1.870	0.907 ±0.030	26.551 ±1.517	0.933 ±0.020	23.942 ±1.656	0.888 ±0.034

Table 3. Ablation of the adversarial component.

	T1	T2	Flair
FID (w/o adv. loss)	51.683	28.643	69.088
Ours	38.057	16.484	36.364

4. CONCLUSION

In this study, we have introduced a novel approach for translating ULF MRI to HF MRI using an adversarial diffusion model with an attention-guided non-diffusive network. Our experiments demonstrated that the proposed approach outperforms the state-of-the-art methods in medical image synthesis. Therefore, this diffusion-based method proves to be a valuable method of improving the quality of ultra-low-field MRI images and presents a promising approach for the synthesis of HF MRI images.

5. ACKNOWLEDGEMENT

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6. COMPLIANCE WITH ETHICAL STANDARDS

This study was performed in line with the principles of the Declaration of Helsinki and approved by the Monash University Human Research Ethics Committee. Informed written consent was obtained from all participants.

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